

Artur Back de Luca

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ABOUT ME

My research focuses on reasoning in neural networks, particularly on which problems they can solve, how they can learn to solve them from data, and how we can certify that they have learned to do so correctly. Understanding these questions is important for building models whose reasoning is more reliable and whose capabilities can be better understood. I have also worked on machine learning on graphs, federated learning, and time-series modeling, including probabilistic forecasting and diffusion-based data generation for foundation models.

EDUCATION

University of Waterloo, Waterloo, Ontario, Canada

Doctor of Philosophy in Computer Science.

Sep 2022 – May 2027 (expected)

Supervisor: Kimon Fountoulakis, Associate Professor

Sapienza University of Rome, Rome, Lazio, Italy

M.Sc. in Artificial Intelligence and Robotics.

Sep 2019 – Jul 2022

Universidade Federal de Santa Catarina, Florianopolis, Santa Catarina, Brazil

B.Sc. in Mechanical Engineering.

Aug 2013 – Feb 2019

RESEARCH EXPERIENCE

Applied Science Intern – **Amazon**, Supply Chain Optimization Technologies (SCOT)

New York, United States

Jun–Sep 2026; Jun–Sep 2025

Support Research Intern – **Huawei Canada**, Noah’s Ark Lab

Toronto, Canada

Feb 2022 – Jan 2023

Research Consultant – **CERTI Foundation** and Micromed

Florianopolis, Brazil

Feb 2021 – Jan 2022

PUBLICATIONS

Conference Proceedings

A. Back de Luca, K. Fountoulakis. Certification from Examples is Hard for Circuits and Transformers under Minimal Overparametrization. To appear in: *Annual Conference on Neural Information Processing Systems* (NeurIPS 2026).

M. Qharabagh, **A. Back de Luca**, G. Giapitzakis and K. Fountoulakis. Learning to Execute Graph Algorithms Exactly with Graph Neural Networks (Spotlight). In: *International Conference on Machine Learning* (ICML 2026).

A. Back de Luca, G. Giapitzakis, and K. Fountoulakis. Learning to Add, Multiply, and Execute Algorithmic Instructions Exactly with Neural Networks. In: *Annual Conference on Neural Information Processing Systems* (NeurIPS 2025).

A. Back de Luca, G. Giapitzakis, S. Yang, P. Veličković, K. Fountoulakis. Positional Attention: Expressivity and Learnability of Algorithmic Computation. In: *International Conference on Machine Learning* (ICML 2025).

A. Back de Luca, K. Fountoulakis. Simulation of Graph Algorithms with Looped Transformers. In: *International Conference on Machine Learning* (ICML 2024).

A. Back de Luca, K. Fountoulakis, S. Yang. Local Graph Clustering with Noisy Labels. In: *International Conference on Learning Representations (ICLR 2024)*.

Workshop Papers and Preprints

G. Giapitzakis, **A. Back de Luca** and K. Fountoulakis. Exact Learning of Permutations for Nonzero Binary Inputs with Logarithmic Training Size and Quadratic Ensemble Complexity. In: High-dimensional Learning Dynamics Workshop at ICML 2025.

A. Back de Luca, G. Zhang, X. Chen, Y. Yu. Mitigating Data Heterogeneity in Federated Learning with Data Augmentation. 2022.

AWARDS

Canada Graduate Research Scholarship (CGRS D)	\$120,000 offered by NSERC.	2026–2028.
President’s Graduate Scholarship (PGS)	\$10,000 offered by the University of Waterloo.	2026–2027.
Graduate Scholarship in Data & AI	\$15,000 offered by TD and Layer 6.	2025–2026.
David R. Cheriton Graduate Scholarship	\$10,000 offered by the University of Waterloo.	2025–2026.
Ontario Graduate Scholarship (OGS)	\$15,000 offered by the Ontario government.	2025–2026.
President’s Graduate Scholarship (PGS)	\$5,000 offered by the University of Waterloo.	2025–2026.
Math Domestic Graduate Award.	\$4,000 offered by University of Waterloo.	2023-2024.
International Master’s Award of Excellence.	\$7,500 offered by University of Waterloo.	2022-2023.

TEACHING ACTIVITIES

University of Waterloo, Teaching Assistant.

Responsibilities include supporting undergraduate course activities, grading assignments and exams, and holding office hours.

CS 338 – Computer Applications in Business: Databases	Winter 2026
CS 235 – Data Types and Structures	Fall 2025
CS 479/679 – Neural Networks	Winter 2025
CS 348 – Introduction to Database Systems	Fall 2024
CS 475/675 – Computational Linear Algebra	Spring 2024
CS 338 – Computer Applications in Business: Databases	Winter 2024
CS 245 – Logic and Computation	Fall 2022, Spring 2023, Fall 2023

OTHER PROFESSIONAL EXPERIENCE

Engineering Intern – NEO Empresarial.	2015 – 2018
Data Analytics Intern – CERTI Foundation.	Summers 2017 and 2018
Research Intern – Fraunhofer Institute for Production Technology.	2017 – 2018
Procurement Intern – Whirlpool/ Embraco.	Summer 2016
Research Intern – Computational Fluid Dynamics Lab (SINMEC).	Spring 2015